Gawtam C R | ED19B007 | PR/22/ED/24/007

EDUCATION

Program	Degree	Institution	% / CGPA	Year
Dual Degree (B.Tech + M.Tech)	Engineering Design, *Specialization - Robotics*	Indian Institute of Technology Madras, Chennai	8.14/10	2024
Class XII	CBSE	Maharishi Vidya Mandir, Chennai	95.4%	2019
Class X	CBSE	Maharishi Vidya Mandir, Chennai	10/10	2017

• Awarded (among **30** out of 800+ students) the IIT Madras **Young Research Fellowship** based on remarkable research potential	2021
• Secured **97.6457 Percentile** in **JEE Advanced** and **99.1046 Percentile** in **JEE Mains** out of over **1.5 Million** aspirants across India	2019
• Secured an All-India Rank of **51** in **UCEED** (Undergraduate Common Entrance Examination for Design) conducted by **IIT Bombay**	2019

RESEARCH AND PROFESSIONAL EXPERIENCE

Systematics *Robotics Desgin Intern* *Dec 2022 – Jun 2023*	
Yaskawa Pvt. Ltd *Robotics Intern* *Jun 2022 – Aug 2022*	*Yaskawa Electric Pvt. Ltd. is an industry leader in Arc welding, Spot welding, Handling, Palletizing and Painting robots.* • Modeled and analyzed three-dimensional robotics palletizing algorithm and mechanics for mixed carton sizes. • Programmed and simulated arc welding and palletizing robots using **MotoSim** EG-VRC[1] and teach pendant. • Developed **a C++ algorithm** to effectively use robot's memory when sending data to the robot from computer.
Young Research Fellow *IIT Madras* *Nov 2022 – Present*	*An illustrious program funded by the IITM batch of '76 to support undergraduate research; Guide - Prof. Satadal Ghosh* • Developing a hybrid algorithm incorporating online obstacle avoidance & path planning for unmanned vehicles • Simulated it in MATLAB environment; **Drafting a research paper**, to be submitted in IEEE Aerospace Journals
Blurgs *Computer vision Intern* *Jun 2021 – Aug 2021*	*Employs real-time vision processing techniques to assist, automate, and augment the capabilities of drones in action* • Analyzed & benchmarked **7+** state of the art **MOT**[2] networks for the startup's **first** product for the **Indian Navy** • Used **FairMOT** to track and get the trail of multiple moving objects and **Re-identify** objects in real-time.
Avishkar Hyperloop *Power Systems Engineer* *Oct 2020 – Jun 2021*	*The leading Asian student competition team with the vision to make Hyperloop transportation a reality in our country* • Designed a batter pack with **high discharge rate** for higher speeds while ensuring safe power supply to the pod • Extended battery pack life by **10%** by reducing peak operational temperature by **10°C** using an Al[3] heat sink.
Imaginate Pvt. Ltd. *Software Dev. Intern* *Apr 2020 – Aug 2020*	*Enables collaboration across different locations using VR/AR for purposes such as industrial design, support and sales.* • Developed a **Unity3D** plugin in **C#** to **upload any 3D model** into a VR using Autodesk **Forge** APIs in **real-time**. • Built a basic **Heroku** app to upload and view any 3D model with **60+** file formats in the web & mobile browser • Achieved **100%** reduction in staff-hours required for this specific task by automating entire process involved.

PROJECTS

Path Finder| PyQt5 *Apr 2021 – May 2021*	• Developed a 2D Path finder from scratch in **python** using state of the art **A-Star** & **Dijkstra** Algorithms in PyQt5 • Designed an **interactive UI** for visualization using **QT Designer;** Documented the project for future use.
Human Powered Segway *Apr 2021 – Jun 2021*	• Lead a **6-member** team involved in ideation of a **novel** mechanism of a fully mechanical alternative of a Segway • Improved efficiency by **50%** by replacing conventional circular-gear pedaling system with **rectilinear pedaling**. • Carried out kinematic analysis to determine **6+ parameters** using MATLAB and SIMULINK to improve efficiency.
CFD[4] Project *Mar 2021 - Apr 2021*	• Designed an **aero foil** for the Front wing of a Formula Student car for maximum **Downforce/Drag** ratio. • Performed **CFD[4] analysis** to determine maximum possible ratio and the optimal setup for better efficiency.
Robotic Arm *Apr 2021 – Jun 2021*	• Designed a small robotic arm with 2 degree of freedom for identifying and lifting objects using electromagnets • Programmed using Arduino mega and used proximity sensors for detection and LED light for sensing color

RELEVANT COURSES AND SKILLS

Courses	Deep Learning, Field and Service robotics, Control Systems, Robotics Lab, Control of Automotive System, Mechatronics System Design, Mechanics and Control of Serial Robots*, Data Structures and Algorithms,	
MOOCs[5]	Algorithmic Toolbox*, Intro to Reinforcement Learning*, Game Development, Money Banking and Finance	
Skills	**Languages:** Python, C, C++, C#	**Tools:** VS Code, MATLAB, LaTeX, MS-Office, Unity3D, Fusion360, Linux
	Frameworks: Tensorflow, Pytorch	**Libraries:** Pandas, NumPy, Scikit Learn, Matplotlib, OpenCV, PyQt5, OpenGL

POSITION OF RESPONSIBILITIES

Department Core, *Placement & Internship Cell 2022*	• **Heading** & mentoring a **3-tier** team in charge of department placement and internship process **1200+** students. • Analyzed past 3 years' list of contacts to eliminate redundant calling in order to increase company conversion. • **Planning & executing** the logistic framework for the first ever shift to hybrid mode of the recruitment process.
Junior Consultant *ShARE 2021*	• Worked in the **'Mobility'** Module, underwent training on focused application of problem-solving in consulting. • Part of a **global team of 700 students** across **15 nations**, enabling firms to transition into '**Do Well, Do Good'**
Coding and Logic Events *Shaastra[4] 2021*	• Handled a total footfall of **300+ teams** and crafted a total of **50+ logical questions** for events in Shaastra'21. • **Ideated and executed 2 brand new** events, Mystery Rooms and E-Contest with **100**+ teams participating.

EXTRA CURRICULAR ACTIVITIES

Drawing	• Won **5+** state level competitions and **10+** District level competitions in drawing (Oil Pastels and Pencils)
Teaching Assistant	• Assisted professors and 20+ students in coursework and grading in the courses GN1102[6] & GN6106[7]
Fusion360 API	• Developed a **python script** to automatically create models in **Fusion360** just by specifying dimensions.
Game Dev. / Gaming	• Love to try out various tools and features in Unity3D; Reached a rank of **Silver 2** in **Valorant** by Riot Games. • Solely developed a **complete 3D Game** in Unity3D with Obstacles, Enemies, Points, UI and Visual Effects.
Sports	• Among the **20 out of 100** students selected for freshman Volleyball training; Part of the Hostel Volleyball team

[1] Motoman Simulator Enhanced Graphics Virtual Robot Controller; [2] Multi Object Tracking; [3] Aluminum; [4] Computational Fluid Dynamics; [5] Massive Open Online Courses [6]-Lifeskills; [7]-Happiness, Habits and Success ** - Upcoming courses; *- Ongoing courses